Bowen Deng

PhD Candidate · AI for Chemistry & Materials

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RESEARCH INTERESTS

Machine learning for chemistry and materials science: learning across molecular representations — 1D sequences (SMILES), 2D molecular graphs, and 3D geometry — with sequence models, graph neural networks, and machine-learned interatomic potentials, for reaction discovery, thermochemistry, and molecular / materials property prediction. Also interested in computer vision (human action recognition) and large language models.

EDUCATION

Ph.D. in Chemistry 2023 – 2026 (expected)
Chimie ParisTech – Université PSL, Paris, France
Advisor: Dr. Thijs Stuyver. Thesis: AI for chemistry & materials (molecular representation learning, MLIPs, thermochemistry, reaction discovery).

M.Sc. in Materials and Chemical Engineering 2020 – 2023
Guangxi University, Nanning, China
Advisor: Prof. Shuangliang Zhao.

B.Eng. in Computer Science and Technology 2016 – 2020
B.Eng. in Materials Science and Engineering dual degree
Southwest University of Science and Technology, Mianyang, China

RESEARCH EXPERIENCE

Research Intern — Computer Vision (Human Action Recognition) 2022
Institute of Automation, Chinese Academy of Sciences (CASIA)
With Dr. Dongchang Liu. Developed a fast MLP-based temporal action detection model (TadML).

PUBLICATIONS

1. Bowen Deng, Thijs Stuyver. MLIP-Enhanced Thermochemistry Predictions across Organic Chemical Space. *Journal of Chemical Theory and Computation*, 22(18), 9526, 2026.
2. Marco Ferrer, Bowen Deng, Javier E. Alfonso-Ramos, Thijs Stuyver. Screening Diels–Alder reaction space to identify candidate reactions for self-healing polymer applications. *Digital Discovery*, 4(11), 3400–3410, 2025.
3. Bowen Deng, Thijs Stuyver. Graph-based deep learning models for thermodynamic property prediction: the interplay between target definition, data distribution, featurization, and model architecture. *Journal of Chemical Information and Modeling*, 65(2), 649–659, 2025.
4. Bowen Deng, P. Chen, P. Xie, Z. Wei, S. Zhao. Iterative machine learning method for screening high-performance catalysts for H_2O_2 production. *Chemical Engineering Science*, 267, 118368, 2023.
5. Bowen Deng, S. Zhao, D. Liu. TadML: A Fast Temporal Action Detection with Mechanics-MLP. *Computer Graphics International Conference (CGI)*, 28–40, 2023.
6. Bowen Deng. Catalysis Distillation Neural Network for the Few-Shot Open Catalyst Challenge. *arXiv / ChemRxiv* (preprint), 2023.
7. Z. Wei, Bowen Deng, P. Chen, T. Zhao, S. Zhao. Palladium-based single atom catalysts for high-performance electrochemical production of hydrogen peroxide. *Chemical Engineering Journal*, 428, 131112, 2022.

AWARDS & HONORS

Kaggle Competitions Silver Medal ×3 — Competitions Expert 2021–2022
Google Brain – Ventilator Pressure Prediction (2021);
Hungry Geese (2021);
Feedback Prize – Evaluating Student Writing (2022)

ACADEMIC SERVICE

Reviewer for *The Journal of Physical Chemistry Letters*, *Digital Discovery*, and ACDSA 2027 (IEEE).

TECHNICAL SKILLS

Programming	Python, C/C++, CUDA, Triton
ML / DL	PyTorch, PyTorch Geometric, e3nn, graph neural networks, machine-learned interatomic potentials, transformers
Cheminformatics	RDKit, SMILES, high-throughput reaction profiling, quasi-RRHO thermochemistry
Comp. chemistry	Gaussian, ORCA, VASP, CP2K, LAMMPS, xTB (DFT / ab initio / MD)
Tooling	Git, Linux, SLURM / HPC clusters
Languages	Chinese (native), English (fluent)